Ayman Osama Hassan

DevSecOps Trainee | Software Engineer (Backend)

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Summary

DevSecOps trainee with hands-on backend and deployment work. Built the asynchronous job pipeline (Celery + Redis) and Docker deployment for a Django backend, and rolled out a maintenance logging system used by 17 technicians across 100 machines. ICPC competitor (3rd at Tanta University).

Technical Skills

- Programming Languages: Python, C++, C, SQL, JavaScript
- Backend & Frameworks: Django, Django REST Framework, Celery
- Infrastructure & Tools: Docker, Docker Compose, Linux CLI, Git, PostgreSQL, Redis, Railway
- In training (NTI): Kubernetes, AWS, Terraform, Ansible, Jenkins CI/CD, SonarQube, Snyk, Prometheus

Education & Training

DevSecOps Engineer Track | NTI, Digital Egypt Youth HireReady — **Oct 2026 – Present**

- 420-hour program: Linux and Bash, Docker, Kubernetes, AWS, OpenStack, Terraform, Ansible, and secure CI/CD pipelines with automated security scanning.

BSc Computer & Automatic Control Engineering | Tanta University | 90% — **2021 – 2026**

Full Stack Web Development Using Python | ITI — **Aug – Sep 2025**

Experience

Maintenance & IT Systems Engineer | Aktan Misr Textile (part-time) — **Aug 2026 – Present**

- Designed and rolled out a maintenance logging system (Google Sheets + Apps Script) used by 17 technicians across 100 machines, tracking breakdowns, downtime, repair time and parts.
- Advise management on ERP selection: reviewed three vendor proposals, questioned vendors on customization, workflows and hosting, and compared Odoo and ERPNext licensing.

Software Projects

Soual: AI Question Generator | Graduation Project | Django, Celery, Redis, Docker

- Containerized the backend with Docker Compose (web, Celery worker, PostgreSQL, Redis) and deployed it on Railway as separate web and worker services for the live defense.
- Built the asynchronous job pipeline: requests queued through Redis and processed by Celery workers calling the AI service, with retries (3 attempts, 30 s apart), batching and a partial-success status.

TCP Chat Application | Python, Sockets, Multithreading

- Built a client/server chat over TCP sockets, with the server handling each client on its own thread.

Engineering Competitions

Competitive Programming | ICPC/ECPC — **2023 – 2025**

- Placed 3rd across Tanta University, one rank from regional qualification; instructor and problem setter for the Tanta University ICPC Community (2025–2026).

Electrical Systems (ECU & Sensors) | T-Racers, IMechE Formula Student 2026 — **2024 – 2026**

- Selected the ECU and sensors, produced the pinout and wrote the electrical documentation; prepared the Project Management materials (team 1st in that event).